

Introduction

ALGEBRIC EXPRESSION

$$a + 4ab - 3c + 7$$

Pronumeral – a symbol or letter which takes the place of a number or a variable

Variable – a quantity which can represent different values

Coefficient – the number which appears before a pronumeral within the same term

coefficient of c is -3

Term – a number and one (or more) pronumeral multiplied together or just a number

Constant – a term with no pronumeral

Solving Equations

Equations

$$s = ut + \frac{1}{2} at^2$$

$$A = \pi r^2$$

$$I = PRT$$

$$s = ut + \frac{1}{2}at^2$$

Non - Calc

Given $s = 62$, $t = 3$ and $a = 10$, find u .

$$62 = u(3) + \frac{1}{2}(10)(3)^2$$

$$62 = 3u + 5(9)$$

$$62 = 3u + 45$$

$$3u = 17$$

$$u = \frac{17}{3}$$

Leave the answer
as improper
fraction.

$$s = \frac{(u+v)}{2} t$$

- Find t given $s = 35$, $u = 10$, $v = 4$

$$35 = \frac{(10+4)}{2} t$$

$$35 = \frac{14}{2} t$$

$$35 = 7t$$

$$t = 5$$

Simplify $\frac{14}{2}$ first

$$A = \pi r^2$$

- Find A when $r = 4.1 \text{ m}$
- Find r when $A = 64 \text{ cm}^2$

use π in your
class pack.

$$A = \pi (4.1)^2$$
$$= 52.81017$$

$$\approx 52.81 \text{ m}^2$$

units given

$$64 = \pi r^2$$

use solve(
function

$$\text{solve}(64 = \pi r^2, r)$$

$$r = 4.5135$$

$$\approx 4.51 \text{ cm}$$

A particular industrial process involves the mining and production of a particular metal to various levels of purity. The production of 1 tonne of the metal of $p\%$ purity costs the company $\$C$ where a good approximation of C is given by the formula:

$$C = 1500 + \frac{125000}{100 - p}$$

- Find (a) the cost of producing one tonne of the metal of 90% purity,
(b) the cost of producing one tonne of the metal of 99% purity.

Note p — not $p\%$

$$\begin{aligned} \text{a) } C &= 1500 + \frac{125000}{100 - 90} \\ &= \$14000 \end{aligned}$$

$$\begin{aligned} \text{b) } C &= 1500 + \frac{125000}{100 - 99} \\ &= \$1265000 \end{aligned}$$

As divers dive to greater depths the pressure on the divers increases due to the weight of the increasing amount of water above them. The pressure, P , in Newtons per square metre, N/m^2 , is given by the formula

$$P = 9.8xd$$

where x metres is how far the diver is below the surface of the water,

and $d \text{ kg/m}^3$ is the density of the water.

Find, in N/m^2 ,

- (a) the pressure at a depth of 5 metres in liquid of density 1000 kg/m^3 ,
- (b) the pressure at a depth of 10 metres in liquid of density 1030 kg/m^3 ,
- (c) the pressure at a depth of 30 metres in liquid of density 1030 kg/m^3 .

$$\begin{aligned} \text{a) } P &= 9.8 (5) (1000) \\ &= 49000000 \text{ N/m}^2 \end{aligned}$$

$$\begin{aligned} \text{b) } P &= 9.8 (10) (1030) \\ &= 100940000 \text{ N/m}^2 \end{aligned}$$